Frank Hu

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Research Interests

Generative AI and foundation models (transformers, large language models, chemical language models, vision language models, diffusion models, flow matching models) for molecule and protein design; gradient-based post-training algorithms for molecular optimization and inverse design problems; physics-informed neural networks for molecular and biomolecular simulations and modeling.

Education

Stanford University — *Stanford, CA*
PhD Candidate in Chemistry, GPA: 4.00/4.00 — August 2022 - Present

Carnegie Mellon University — *Pittsburgh, PA*
MS in Chemistry, GPA: 3.97/4.00 — August 2020 - August 2022
BS in Chemistry, GPA: 3.97/4.00 — August 2018 - August 2022

Research

Markland Lab and Rotskoff Lab, Department of Chemistry, Stanford University — *Stanford, CA*
Graduate researcher with Professor Thomas E. Markland and Professor Grant M. Rotskoff — August 2022 - Present

- Building foundation chemical language models and machine learning frameworks for small-molecule design, optimization, and conformational sampling
- Building chemical language model-based frameworks for direct interpretation of spectroscopic data
- Developing data-efficient machine-learned force fields for condensed phase molecular dynamics

Prescient Design/AI4DD, Genentech — *South San Francisco, CA*
Machine learning research intern with the foundation models team — June 2026 - September 2026

- Developed large-scale asynchronous reinforcement learning for small-molecule design with open-weight LLMs
- Assessed the sample efficiency of frontier LLMs in continuous and discrete batch optimization settings

Yaron Lab, Department of Chemistry, Carnegie Mellon University — *Pittsburgh, PA*
Undergraduate and Master's researcher with Professor David J. Yaron — August 2018 - August 2023

- Built a novel framework for machine-learned parameterization of semiempirical chemical Hamiltonians
- Thesis title: *DFTBML: A Machine-Learned Density Functional Based Tight Binding Model for the Deep Learning of Chemical Hamiltonians*

Jiang Lab, Department of Mathematics and Statistics, Georgia State University — *Atlanta, GA*
Research intern with Professor Yi Jiang — June 2017 - April 2018

- Developed computational methods for analyzing cancer cell metastasis based on microscopy image data

Publications and Presentations

Publications

1 A. Aneesh, S. Ibarraran, **F. Hu**, J. Liu, G. M. Rotskoff. Distillation, Self-Training, and the Fragility of Learned Chemical Reasoning. *arXiv preprint, arXiv:2610.24199* (2026)

2 J. Liu, S. Ibarraran, **F. Hu**, S. Yang, X. A. Feng, A. Park, A. Aneesh, L. Bintu, A. Dunn, G. M. Rotskoff. Generative modeling of intrinsically disordered protein regions by reinforcing sparse autoencoder features. *arXiv preprint, arXiv:2610.02189* (2026)

3 S. Dunne, S. Ibarraran, **F. Hu**. SIGMol: Enabling Molecular Design with Flow Map Chemical Language Models. *NeurIPS 2026 Workshop on Beyond Next Token Prediction: Diffusion and Flow Models for Next-Generation Decoding* (2026)

4 **F. Hu***, S. Chennakesavalu*, Z. Wang, P. Suriana, B. Vani, K. Shmilovich, K. Chuang, C. Grambow. Training Large Language Models for Small-Molecule Design with Synthetic Task Scaling. *arXiv preprint, arXiv:2609.04735* (2026)

5 **F. Hu**, S. Chennakesavalu, D. Graff. Frontier LLMs are effective batch optimizers: Assessing reasoning models in continuous and discrete settings. *arXiv preprint, arXiv:2609.03177* (2026)

6 S. Ibarraran, S. Chennakesavalu, **F. Hu**, G. M. Rotskoff. Efficient, Few-Shot Directed Evolution with Energy Rank Alignment. *J. Chem. Inf. Model.*, **66**, 17, 10609–10621 (2026)

7 N. I. Hausman, J. Kelly, M. S. Chen, **F. Hu**, A. Lee, A. Montoya-Castillo, G. S. Schlau-Cohen, T. E. Markland. Streamlining analysis and design of two-dimensional electronic spectroscopy using machine learning. *J. Chem. Phys.*, **165**, 074109 (2026)

8 J. Liu, S. Ibarraran*, **F. Hu***, A. Park*, A. Dunn, G. M. Rotskoff. Generative design of intrinsically disordered protein regions with IDiom. *ICML 2026 Workshop on Generative and Agentic AI for Biology* (2026)

9 J. K. Rasch, A. L. Clayborn, D. Sole-Barber, S. Wan, C. A. Saavedra, A. J. Fairhall, **F. Hu**, J. Bartz, Y. Podorova, A. J. Bergsten, Z. D. Young, I. R. Barrera, T. H. K. Mone, N. Asgari, T. E. Markland, R. H. Goldsmith. Label-Free Optical Differentiation of Single Diffusing Amino Acids at Picomolar Concentrations. *Under review, Nature Nanotechnology* (2026)

10 S. Ibarraran, S. Chennakesavalu, **F. Hu**, G. M. Rotskoff. Efficient, Few-shot Directed Evolution with Energy Rank Alignment. *ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design* (2026)

11 **F. Hu**, J. M. Tubb, D. Argyropoulos, S. Golotvin, M. Elyashberg, G. M. Rotskoff, M. W. Kanan, T. E. Markland. Pushing the Limits of One-Dimensional NMR Spectroscopy for Automated Structure Elucidation Using Artificial Intelligence. *J. Chem. Inf. Model.*, **66**, 12, 6992–7002 (2026)

12 S. Chennakesavalu, **F. Hu**, S. Ibarraran, G. M. Rotskoff. Aligning Transformers with Continuous Feedback via Energy Rank Alignment. *Conference on Neural Information Processing Systems* (2025)

13 J. Kelly, **F. Hu**, A. Damiani, M. S. Chen, A. Snider, M. Son, A. Lee, P. Gupta, A. Montoya-Castillo, T. J. Zuehlsdorff, G. S. Schlau-Cohen, C. M. Isborn, T. E. Markland. Two-Dimensional Electronic Spectroscopy in the Condensed Phase Using Equivariant Transformer Accelerated Molecular Dynamics Simulations. *J. Phys. Chem. Lett.*, **16**, 22, 5561–5569 (2025)

14 S. Chennakesavalu, **F. Hu**, S. Ibarraran, G. M. Rotskoff. Aligning Transformers with Continuous Feedback via Energy Rank Alignment. *ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design* (2025)

15 **F. Hu**, M. S. Chen, G. M. Rotskoff, M. W. Kanan, T. E. Markland. Accurate and Efficient Structure Elucidation from Routine One-Dimensional NMR Spectra Using Multitask Machine Learning. *ACS Cent. Sci.*, **10**, 11, 2162–2170 (2024)

16 S. Chennakesavalu, S. K. Manikandan, **F. Hu**, G. M. Rotskoff. Adaptive nonequilibrium design of actin-based metamaterials: Fundamental and practical limits of control. *Proc. Natl. Acad. Sci.*, **121**, 8, e2310238121 (2024)

17 P. Eastman, R. Galvelis, R. P. Peláez, C. R. A. Abreu, S. E. Farr, E. Gallicchio, A. Gorenko, M. M. Henry, **F. Hu**, J. Huang, A. Krämer, J. Michel, J. A. Mitchell, V. S. Pande, J. PGLM Rodrigues, J. Rodriguez-Guerra, A. C. Simmonett, S. Singh, J. Swails, P. Turner, Y. Wang, I. Zhang, J. D. Chodera, G. De Fabritiis, T. E. Markland. OpenMM 8: Molecular Dynamics Simulation with Machine Learning Potentials. *J. Phys. Chem. B*, **128**, 1, 109–116 (2024)

18 **F. Hu**, F. He, D. J. Yaron. Treating Semiempirical Hamiltonians as Flexible Machine Learning Models Yields Accurate and Interpretable Results. *J. Chem. Theory Comput.*, **19**, 18, 6185–6196 (2023)

Presentations

1 **(Contributed talk) Climbing the Combinatorial Mountain: Using Artificial Intelligence to Push the Limits of 1D NMR for Structure Elucidation**. SMASH 2026, Breckenridge, CO

2 **(Poster) Generative design of intrinsically disordered protein regions with IDiom**. ICML 2026 Workshop on Generative and Agentic AI for Biology, Seoul, South Korea

3 **(Poster) Efficient, Few-shot Directed Evolution with Energy Rank Alignment**. ICLR 2026 Workshop on Generative and Experimental Perspectives for Biomolecular Design, Rio de Janeiro, Brazil

4 **(Invited talk) Using AI to push the limits of 1D NMR for structure elucidation**. Novartis Graduate Colloquium 2026, Novartis, Cambridge, MA

5 **(Poster) Pushing the Limits of One-Dimensional NMR Spectroscopy for Automated Structure Elucidation Using Artificial Intelligence**. Berkeley Statistical Mechanics Meeting 2026, University of California, Berkeley, CA

6 **(Poster) Aligning Transformers with Continuous Feedback via Energy Rank Alignment**. Conference on Neural Information Processing Systems (NeurIPS) 2025, San Diego, CA

7 **(Poster) Accurate and Efficient Structure Elucidation from NMR Using Artificial Intelligence**. SMASH 2025, Porto, Portugal

8 **(Contributed talk) Automated structure elucidation from NMR using artificial intelligence**. Young WATOC (World Association of Theoretical and Computational Chemists) 2025, Oslo, Norway

9 **(Poster) Energy Rank Alignment**. ICLR 2025 Workshop on Generative and Experimental Perspectives for Biomolecular Design, Singapore

10 **(Poster) Accurate and Efficient Structure Elucidation from Routine One-Dimensional NMR Spectra Using Multitask Machine Learning**. Experimental Nuclear Magnetic Resonance Conference 2025, Pacific Grove, CA

11 **(Poster) Structure elucidation from NMR using machine learning**. Berkeley Statistical Mechanics Meeting 2025, University of California, Berkeley, CA

12 **(Poster) Accurate and Efficient Structure Elucidation from Routine One-Dimensional NMR Spectra Using Multitask Machine Learning**. American Conference on Theoretical Chemistry 2024, University of North Carolina, Chapel Hill, NC

13 **(Poster) Accurate and Efficient Structure Elucidation from Routine One-Dimensional NMR Spectra Using Multitask Machine Learning**. West Coast Theory Meeting 2024, University of California, Merced, CA

14 **(Contributed talk) Semiempirical Hamiltonians as Flexible Model Forms for Machine Learning**. ACS 2023, San Francisco, CA

15 **(Poster) DFTBML: A Machine-Learned Density Functional Based Tight Binding Model for the Deep Learning of Chemical Hamiltonians and Properties**. West Coast Theory Meeting 2023, University of California, Davis, CA

16 **(Contributed talk) DFTBML: A Machine-Learned Density Functional Based Tight Binding Model for the Deep Learning of Chemical Hamiltonians and Properties**. Beckman Symposium 2022, Rutgers University, NJ

Honors and Awards

John Stauffer Memorial Fellowship *2026*
Stanford University, Stanford, CA

Center for Molecular Analysis and Design (CMAD) Fellowship *2024 - 2026*
Stanford University, Stanford, CA

Poster Prize *2025*
SMASH 2025, Porto, Portugal

Poster Prize *2024*
West Coast Theory Meeting 2024, University of California, Merced, CA

Poster Prize *2023*
West Coast Theory Meeting 2023, University of California, Davis, CA

NSF GRFP Honorable Mention *2023*
National Science Foundation

College Chemistry Award *2022*
Society of Analytical Chemists of Pittsburgh

Beckman Scholar *2021 - 2022*
Arnold and Mabel Beckman Foundation

National Merit Scholar *2018 - 2022*
National Merit Scholarship Corporation

Leadership and Service

Stanford Graduate Student Affairs Committee Industry Partnerships Co-Chair *July 2025 - March 2026*
Stanford University, Stanford, CA

◦ Networking and outreach with industry representatives to boost campus recruiting for chemistry PhD students

Stanford FAST Mentor *October 2024 - March 2026*
Stanford University, Stanford, CA

◦ Scientific outreach work and mentoring underprivileged high school students doing science fair projects

Teaching Assistant *August 2022 - May 2023, January 2025 - March 2025*
Department of Chemistry, Stanford University, Stanford, CA

◦ TA for 4 quarters: general chemistry, statistical mechanics, quantum chemistry, and computational chemistry

Head Teaching Assistant *August 2021 - August 2022*
School of Computer Science, Carnegie Mellon University, Pittsburgh, PA

◦ Head TA for an introductory computer science course

◦ Developed course assignments and exams and arranged TA staff logistics in addition to teaching